

Sampling Distribution of the Mean

Probability & Statistics (MT2005)

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What is a Sampling Distribution?

- A sampling distribution is the probability distribution of a statistic (like the mean, variance etc.) based on many samples taken from the same population.
- Think of it this way:
 - You take many random samples of size n from a population.
 - You calculate the mean of each sample.
 - Then you plot all those means $\rightarrow$ this gives you the sampling distribution of the sample mean.

Basic Terminologies of Sampling Distribution

- **Population:** The complete set of individuals or items under study.
- **Sample:** A subset of the population, selected for analysis.
- **Statistic:** A numerical value calculated from a sample (e.g., sample mean $\bar{X}$, sample variance S^2).
- **Parameter:** A numerical value describing a characteristic of the population (e.g., population mean μ , population variance σ^2).
- **Sampling Distribution:** The probability distribution of a given statistic based on a random sample.
- **Standard Error (SE):** The standard deviation of a sampling distribution; measures variability of a statistic.
- **Bias:** The difference between the expected value of a statistic and the true value of the parameter.
- **Law of Large Numbers:** As sample size increases, the sample mean approaches the population mean.

Example: Sampling Distribution of the Mean

Population: $\{4, 6, 8\}$

Sample size: 2 (with replacement)

- All possible samples:

$(4, 4), (4, 6), (4, 8), (6, 4), (6, 6), (6, 8), (8, 4), (8, 6), (8, 8)$

Step 2: Calculate Sample Means

Sample	Sample Mean
(4, 4)	4
(4, 6)	5
(4, 8)	6
(6, 4)	5
(6, 6)	6
(6, 8)	7
(8, 4)	6
(8, 6)	7
(8, 8)	8

Step 3: Frequency Distribution of Sample Means

Sample Mean	Frequency	Probability
4	1	$\frac{1}{10}$
5	2	$\frac{2}{10}$
6	3	$\frac{3}{10}$
7	2	$\frac{2}{10}$
8	1	$\frac{1}{10}$

Step 4: Mean and Variance of Sampling Distribution

Mean:

$$\mu_{\bar{X}} = 4 \left(\frac{1}{9} \right) + 5 \left(\frac{2}{9} \right) + 6 \left(\frac{3}{9} \right) + 7 \left(\frac{2}{9} \right) + 8 \left(\frac{1}{9} \right) = 6$$

Variance:

$$E(\bar{X}^2) = \frac{336}{9} = 37.333$$

$$\sigma_{\bar{X}}^2 = \text{Var}(\bar{X}) = 37.333 - (6)^2 = 1.333$$

Sampling Distribution of Difference of Means

Population A: $\{2, 4\}$ Sample size: 2

Population B: $\{5, 7\}$ Sample size: 2

Possible sample means:

- From A: 2, 3, 3, 4
- From B: 5, 6, 6, 7

Rectangular Table of Differences

$\bar{X}_A \setminus \bar{X}_B$	5	6	6	7
2	-3	-4	-4	-5
3	-2	-3	-3	-4
3	-2	-3	-3	-4
4	-1	-2	-2	-3

Summary of Differences

- Possible differences: $-5, -4, -3, -2, -1$
- Frequency table with probabilities:

Difference	Frequency	Probability
-5	1	$\frac{1}{16}$
-4	4	$\frac{4}{16} = \frac{1}{4}$
-3	6	$\frac{6}{16} = \frac{3}{8}$
-2	4	$\frac{4}{16} = \frac{1}{4}$
-1	1	$\frac{1}{16}$

- Most common difference: -3

Mean and Variance of Sampling Distribution

Step 1: Find the Mean (Expected Value)

$$\begin{aligned} E(\bar{X}_A - \bar{X}_B) &= \sum (\text{value}) \times (\text{probability}) \\ &= (-5) \left(\frac{1}{16} \right) + (-4) \left(\frac{4}{16} \right) + (-3) \left(\frac{6}{16} \right) + (-2) \left(\frac{4}{16} \right) + (-1) \left(\frac{1}{16} \right) \\ &= \frac{-5 - 16 - 18 - 8 - 1}{16} = \frac{-48}{16} = -3 \end{aligned}$$

Step 2: Find the Variance

First calculate $E[(\bar{X}_A - \bar{X}_B)^2]$:

$$\begin{aligned} &= (-5)^2 \left(\frac{1}{16} \right) + (-4)^2 \left(\frac{4}{16} \right) + (-3)^2 \left(\frac{6}{16} \right) + (-2)^2 \left(\frac{4}{16} \right) + (-1)^2 \left(\frac{1}{16} \right) \\ &= \frac{25 + 64 + 54 + 16 + 1}{16} = \frac{160}{16} = 10 \end{aligned}$$

Now, variance:

$$\begin{aligned}\sigma_{\bar{X}_A - \bar{X}_B}^2 &= \text{Var}(\bar{X}_A - \bar{X}_B) = E[(\bar{X}_A - \bar{X}_B)^2] - (E(\bar{X}_A - \bar{X}_B))^2 \\ &= 10 - (-3)^2 = 10 - 9 = 1\end{aligned}$$

Properties of Sampling Distribution of Mean

Sampling Distribution of Mean:

- **Mean:** The mean of the sampling distribution is the population mean, $\mu_{\bar{X}} = \mu$.
- **Variance:** The variance of the sampling distribution is $\text{Var}(\bar{X}) = \frac{\sigma^2}{n}$, where σ is the population standard deviation.
- **Standard Error:** The standard error is $\text{SE} = \frac{\sigma}{\sqrt{n}}$.
- **Shape (for large n):** The sampling distribution approximates a normal distribution as n increases.
- **Central Limit Theorem:** The Central Limit Theorem holds for all sample sizes when the population is normal or for large n .

Properties of Sampling Distribution Difference of Means

Sampling Distribution of Difference of Means:

- **Mean:** The mean of the sampling distribution of the difference of means is the difference of population means, $\mu_{\bar{X}_A - \bar{X}_B} = \mu_A - \mu_B$.
- **Variance:** The variance of the sampling distribution of the difference of means is $\text{Var}(\bar{X}_A - \bar{X}_B) = \frac{\sigma_A^2}{n_A} + \frac{\sigma_B^2}{n_B}$.
- **Standard Error:** The standard error is $\text{SE}(\bar{X}_A - \bar{X}_B) = \sqrt{\frac{\sigma_A^2}{n_A} + \frac{\sigma_B^2}{n_B}}$.
- **Shape (for large n):** The sampling distribution of the difference of means approximates a normal distribution as n increases.
- **Central Limit Theorem:** The Central Limit Theorem holds for all sample sizes when the populations are normal or for large n .

Proof Outline for Few Properties

Property 1: Mean of $\bar{X}$

$$E(\bar{X}) = \mu$$

Proof:

$$\begin{aligned}\bar{X} &= \frac{1}{n} \sum_{i=1}^n X_i \\ E(\bar{X}) &= E\left(\frac{1}{n} \sum_{i=1}^n X_i\right) = \frac{1}{n} \sum_{i=1}^n E(X_i) \\ &= \frac{1}{n} \cdot n \cdot \mu = \mu\end{aligned}$$

Proof Outline for Few Properties Cont'd...

Property 2: Variance and Standard Error

$$\text{Var}(\bar{X}) = \frac{\sigma^2}{n}, \quad SE(\bar{X}) = \frac{\sigma}{\sqrt{n}}$$

Proof:

$$\begin{aligned}\text{Var}(\bar{X}) &= \text{Var}\left(\frac{1}{n} \sum_{i=1}^n X_i\right) \\ &= \frac{1}{n^2} \sum_{i=1}^n \text{Var}(X_i) \\ &= \frac{1}{n^2} \cdot n \cdot \sigma^2 = \frac{\sigma^2}{n}\end{aligned}$$

Property 3: Shape of Distribution

- If population is normal, $\bar{X}$ is normal for any n .
- If population is not normal:
 - For large n (≥ 30), $\bar{X}$ is approximately normal.

Central Limit Theorem

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Central Limit Theorem (CLT)

Statement: Let $X_1, X_2, \dots, X_n$ be i.i.d. with mean μ and variance σ^2 .
Then:

$$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \rightarrow N(0, 1) \text{ as } n \rightarrow \infty$$

CLT: Proof Sketch (MGF)

Define:

$$Y_i = \frac{X_i - \mu}{\sigma}, \quad Z_n = \frac{1}{\sqrt{n}} \sum_{i=1}^n Y_i$$

- $E(Y_i) = 0$, $\text{Var}(Y_i) = 1$
- Z_n is a sum of i.i.d with mean 0 and variance 1
- As $n \rightarrow \infty$, MGF of $Z_n \rightarrow e^{t^2/2}$ (MGF of $N(0, 1)$)

Therefore, $Z_n \rightarrow N(0, 1)$ in distribution.

Conclusion

- Sample mean $\bar{X}$ has expected value μ
- Standard error decreases as n increases
- CLT allows us to assume normality for large n even if the population is not normal

Central Limit Theorem Examples

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CLT: One Population – Problem

Given: Time to complete a quiz is normally distributed with:

- Mean $\mu = 30$ minutes
- Standard deviation $\sigma = 8$ minutes
- Sample size $n = 36$

Tasks:

- 1 Find $P(\bar{X} < 28)$
- 2 Find $P(29 < \bar{X} < 31)$

CLT: One Population – Solution

Step 1: Sampling distribution

$$\sigma_{\bar{X}} = \frac{\sigma}{\sqrt{n}} = \frac{8}{6} = 1.33$$

Part (i):

$$Z = \frac{28 - 30}{1.33} \approx -1.50 \Rightarrow P(Z < -1.50) \approx \boxed{0.0668}$$

Part (ii):

$$Z_1 = -0.75, \quad Z_2 = 0.75$$

$$P(-0.75 < Z < 0.75) = 0.7734 - 0.2266 = \boxed{0.5468}$$

CLT: Two Populations – Problem

Given:

- School A: $\mu_A = 75, \sigma_A = 10, n = 40$
- School B: $\mu_B = 70, \sigma_B = 12, m = 40$

Tasks:

- 1 Find $P(\bar{X}_A - \bar{X}_B > 8)$
- 2 Find $P(2 < \bar{X}_A - \bar{X}_B < 5)$

CLT: Two Populations – Solution

$$\mu_D = \mu_A - \mu_B = 5, \quad \sigma_D = \sqrt{\frac{100}{40} + \frac{144}{40}} = \sqrt{6.1} \approx 2.47$$

Part (i):

$$Z = \frac{8 - 5}{2.47} \approx 1.215 \Rightarrow P(Z > 1.215) \approx \boxed{0.1114}$$

Part (ii):

$$Z_1 = -1.215, \quad Z_2 = 0$$

$$P(-1.215 < Z < 0) = 0.5 - 0.1114 = \boxed{0.3886}$$